Survival Analysis

MODIFIED
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WEBR STATUS
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Data considerations

- Similar to when the outcome is binary, survival outcomes should be coded using 0's and 1's where 1 indicates the occurrence of an event and 0 otherwise.
- Similar to how we encode censoring variables, we consider the outcome to be degenerate. Meaning that once an observation experiences an outcome, all future outcome variables should also be coded with a 1 ("last-observation-carried-forward").

Point-treatment survival problems

Up to this point, we've been ignoring that the `covid` dataset should be treated as a point-treatment survival problem. Let's re-estimate the effect of the randomized treatment with a survival framework.

- We need to transform the data from long to wide format
- and impute the outcome using last-observation-carried-forward.

Our modified dataset should look like this:

	Baseline confounders		Time-dependent confounders		Trt.	Censoring	Survival				
ID	W1	W2	$L1_1$	$L2_2$	A	C_1	Y_2	C_2	Y_3	...	Y_{T+1}
1	50	0	1012	23	0	1	0	1	0	...	1
2	68	1	25300	17	0	1	1	NA	1	...	1
3	32	1	255	95	1	1	0	1	0	...	0
...	...	...	...	...	...	...	...	...	...	...	...
N	79	0	6783	13	0	0	NA	NA	NA	...	NA

Structure of data with survival outcome and point-treatment. Adapted from Hoffman et al., 2022.

Installing package ...

```
library(dplyr)
library(tidyr)

covid <- pivot_wider(
  covid,
  values_from = "event",
  names_from = "days",
  names_prefix = "day_",
  names_sort = TRUE
)

covid <-
  event_locf(covid, paste0("day_", 1:15)) |>
  mutate(across(starts_with("day_"), ~ replace_na(.x, 0))) |>
  as.data.frame()

head(covid)
```

> Use the function `event_locf()` to make sure the outcome variables are correctly recorded.

- Instead of just estimating the effect of treatment on the outcome at the last time point, we can estimate the effect of a treatment on an outcome at all follow-up intervals.

Let's estimate the effect of the treatment on intubation at each day. To do so, we can use the function `lmtp_survival()` .

Installing package ...

```
W <- c("age", "sex", "bmi", "smoke", "o2",
```

```
         "num_comorbid", "num_symptoms", "bilat", "dyspnea", "hyper")

fit <- lmtp_survival(
  data = covid,
  trt = "A",
  outcomes = grep("^day", names(covid), value = TRUE),
  baseline = W,
  shift = static_binary_on,
  folds = 1,
  estimator = "lmtp_sdr",
  control = lmtp_control(.learners_outcome_folds = 2,
                         .learners_trt_folds = 2)
)
```

We can now visualize our results using a survival plot.

> The main result of an `lmtp` object can be extracted using the `tidy()` function from the `broom` package.

Installing package ...

```
library(ggplot2)

results <- tidy(fit)

ggplot(results, aes(x = time, y = estimate)) +
  geom_step() +
  labs(x = "Days", y = "Survival probability") +
  scale_y_continuous(limits = c(0.65, 1),
                     n.breaks = 10,
                     expand = c(0, 0)) +
  scale_x_continuous(limits = c(1, 15),
                     n.breaks = 15,
                     expand = c(0.01, 0))
```

Time-varying treatment

Hoffman et al. (2024) demonstrated the use of modified treatment policies for survival outcomes to assess the effect of delaying invasive mechanical ventilation (IMV) on mortality among patients hospitalized with COVID-19 in New York City during the first COVID-19 wave. A synthetic version of the data used for that analysis has been loaded into R as `intubation`.

Censoring

Trt.

Survival

Baseline confounders

Time-dependent confounders

ID	W1	W2	$L1_1$	$L2_2$	A_1	C_1	Y_2	$L1_2$	$L2_2$	A_2	C_2	Y_3	...	Y_{T+1}
1	50	0	1012	23	0	1	0	1901	23	0	1	0	...	1
2	68	1	25300	17	0	1	1	NA	NA	NA	NA	1	...	1
3	32	1	255	95	1	1	0	270	52	1	1	0	...	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
N	79	0	6783	13	0	0	NA	NA	NA	NA	NA	NA	...	NA

First time point *Second time point*

Structure of data with survival outcome and time-varying treatment. Adapted from Hoffman et al., 2022.

- The data consists of $n = 2000$ observations hospitalized with COVID-19 and who were followed for $\tau = 14$ days.
- There are 10 baseline confounders and 4 time-varying confounders.
- The outcome of interest is an indicator for death on day t.
- Observations are subject to loss-to-follow-up due to either hospital discharge or transfer.

age	sex	bmi	cad	home_o2_yn	dm	htn	cva	cirrhosis	asthma	copd	active_cancer	C_00	C_01	C_02	C_03	C_04	C_05	C_06	C_07
33	1	25.2	0	1	0	0	0	0	0	0	0	1	1	0					
63	1	32.46	0	1	0	1	0	0	0	0	0	1	1	1	1	1			
46	1	44.9	0	0	1	1	0	0	0	0	0	1	1	1	1	1	1	1	1
75	1	23.77	0	0	0	1	0	0	0	0	0	1	1	1	1	1	1	0	
36	0	27.4	0	0	0	0	1	0	0	0	0	1	1	1	1	1			
65	1	23	0	0	1	0	0	0	0	0	0	1	1	1	1				
97	0	21	0	0	0	1	0	0	0	0	0	1	1	1	1	1	1	1	1
24	0	39.9	0	0	0	0	0	0	0	0	0	1	1						
73	0	31.8	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
50	1	31	0	0	0	1	0	0	1	0	0	1	1	1	1	1	1	1	1

Previous 1 2 3 4 5 ... 200 Next

Let's consider the following intervention

$$\mathsf{d}_t(a_t, h_t) = \begin{cases} 1 \text{ if } a_t = 2 \\ a_t \text{ otherwise,} \end{cases}$$

where A_t is a 3-level categorical variable: 0, no supplemental oxygen; 1, non-IMV supplemental oxygen support; 2, IMV.

In words, this function corresponds to an intervention where patients who were naturally observed as receiving IMV on day t instead receive non-IMV supplemental oxygen. Let's translate this policy to an R function that we can use with `lmtp`.

Installing package ...

```
d <- function(data, trt) {
  ifelse(data[[trt]] == "2", factor("1", levels = 0:2), data[[trt]])
}
```

We can now estimate the effect of delaying IMV by 1-day on 14-day mortality.

Installing package ...

```
day <- 14
intubated <- paste0("I_", sprintf("%02d", 0:(day - 1)))
confounders <- names(intubation)[1:10]
death <- paste0("Y_", sprintf("%02d", 1:day))
discharge <- paste0("C_", sprintf("%02d", 0:(day - 1)))
timevary <- grep("^L_", names(intubation), value = TRUE)
timevary <- lapply(0:(day - 1), \(x) timevary[grepl(sprintf("%02d", x), timevary)

lmtp_sdr(
  intubation,
  trt = intubated,
  outcome = death,
  baseline = confounders,
  time_vary = timevary,
  cens = discharge,
  shift = d,
  mtp = TRUE,
  outcome_type = "survival",
  folds = 1,
  k = 2
)
```

Competing risks

In the context of survival analysis, competing risks refer to events that preclude the occurrence of the primary event of interest. In the previous example, we treated hospital discharge or transfer as censoring events. These events, however, are actually competing risks because we know that if a patients was discharged or transferred out of the ICU on day t they didn't die on that day.

> Treating competing risks as censoring events involves estimating the effect of an intervention that eliminates the competing event. The identification assumptions for this intervention are stronger than intervention that doesn't consider eliminating the competing event.

In the presence of competing risks, `lmtp` can estimate cumulative incidence effects. Cumulative incidence effect can be interpreted as the *total effect* of treatment operating through pathways that include the competing events. Let's re-evaluate the effect of the previous intervention on 14-day mortality but instead treat discharge or transfer as a competing risk. We first need to modify the data:

- Flip the columns corresponding to discharge or transfer so that a 1 indicates a discharge or transfer occurred
- Impute missing values for discharge and death using last-observation carried forward because they are both deterministic variables once they have occurred (i.e., if a patient died, their probability of discharge or transfer is 0 and vice-versa)

Installing package ...

```
# Function to impute missing values with the last observed observation
locf <- function(data, columns) {
  for (col in columns) {
    # Get the column index
    index <- which(names(data) == col)

    # Skip the first column
    if (index == 1) {
      next
    }

    # Find rows with values
    na_rows <- is.na(data[[index]])
    # Replace NA values with the value from the column to the left
    data[[index]][na_rows] <- data[[index - 1]][na_rows]
  }

  return(data)
}

intubation <-
  mutate(intubation, across(starts_with("C_"), function(x) {
    case_when(x == 1 ~ 0, x == 0 ~ 1, TRUE ~ x)
  })) |>
  locf(discharge) |>
  locf(death) |>
  mutate(across(all_of(death), \(x) ifelse(is.na(x), 0, x)))
```

We can now re-estimate the effect of the intervention. Instead of passing the `discharge` vector to the `cens` argument, we pass it to the `compete` argument.

Installing package ...

```
lmtp_sdr(
  intubation,
  trt = intubated,
  outcome = death,
  baseline = confounders,
  time_vary = timevary,
  compete = discharge,
  shift = d,
  mtp = TRUE,
  outcome_type = "survival",
  folds = 1,
  k = 2
)
```

References

Díaz, Iván, Katherine L Hoffman, and Nima S Hejazi. 2024. "Causal Survival Analysis Under Competing Risks Using Longitudinal Modified Treatment Policies." *Lifetime Data Analysis* 30 (1): 213–36.

Díaz, Iván, Nicholas Williams, Katherine L Hoffman, and Edward J Schenck. 2023. "Nonparametric Causal Effects Based on Longitudinal Modified Treatment Policies." *Journal of the American Statistical Association* 118 (542): 846–57.

Hoffman, Katherine L., Diego Salazar-Barreto, Nicholas Williams, Kara E. Rudolph, and Ivan Diaz. 2024. "Studying Continuous, Time-Varying, and/or Complex Exposures Using Longitudinal Modified Treatment Policies." https://arxiv.org/abs/2304.09460.

Young, Jessica G, Mats J Stensrud, Eric J Tchetgen Tchetgen, and Miguel A Hernán. 2020. "A Causal Framework for Classical Statistical Estimands in Failure-Time Settings with Competing Events." *Statistics in Medicine* 39 (8): 1199–1236.